# Hakancan Ozturk

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## PROFESSIONAL EXPERIENCE

**Software Development Engineer** | ***Amazon*** *- London, UK* 01/2025 - Present

- Migrating Prime Video UI elements from React to SolidJS to enhance performance, achieving 10% faster app UI for 100M+ customers through LLM-powered automation pipelines for efficient component refactoring.
- Developed LLM agents with LangChain connected via MCP to perform real-time analysis on over 10 million Prime Video operational metrics, enabling rapid detection of system regressions (Project won London-wide hackathon).
- Productionizing the award-winning diagnostic system using AWS Step Functions, Lambda, S3, and LangFuse for agent evaluation, reducing oncall investigation time by 70% during critical Prime Video outages.

**Founding Engineer** | ***Albus Technologies*** *- London, UK* 05/2024 - 01/2025

- Engineered and optimized a scalable Retrieval-Augmented Generation (RAG) system with context-enriched vector search, boosting retrieval relevance by 30% and serving 100K+ customers.
- Developed scalable real-time Document Semantic Extraction pipelines and LLM agents using AWS (Lambda, S3, SQS, EC2), processing millions of PDF pages and over 50K minutes of audio.
- Established automated CI/CD workflows (Docker, GitHub Actions) to deploy FastAPI and Lambda endpoints, enabling dozens of concurrent file uploads with low-latency, robust production.

**Computing Researcher** | ***Max Planck Institute*** *- Stuttgart, Germany* 06/2022 - 12/2022

- Enhanced robotics dynamics prediction by deploying ML data pipelines (SVMs, curve fits) analyzing 10TB simulation data; optimized CFD simulations via novel HPC scheduling, achieving 200x speedup, enabling 3 publications.

## EDUCATION

**Imperial College London** | *MSc in Applied Computational Science and Engineering* 09/2023-09/2024
Highest overall grade in class | Class representative Distinction (78.27%)
*Modules:* Machine/Deep Learning, Numerical Methods, Computational Maths, Optimization

**Koc University** | *BSc in Mechanical Engineering* 2020-2023
Ranked 1st in class | Graduated one year early | Merit scholarship ($30k annually) GPA: 3.99/4.00

## PROJECTS

**MSc Dissertation: AI Surrogate Modeling for Turbulent Flow Simulations** | *Imperial College London* 2024 - 2025

- Discovered a novel Grid-Invariant AI architecture (PyTorch) combining convolutional autoencoders and adversarial networks to simulate high-fidelity turbulent flows, achieving unprecedented grid independence and scalability.
- Conducted 2000+ GPU hours of High-Performance Computing (HPC) for model optimization, enhancing long-term stability by 35% and prediction accuracy by 50%.
- Actively open-sourcing advancements (manuscripts in prep/review), with project development backed by NVIDIA.

## SKILLS & PUBLICATIONS

Languages & Libraries: Python, TypeScript, C++, MATLAB, PyTorch, scikit-learn, pandas, numpy, FastAPI, React
Cloud & DevOps: Docker, AWS, Github Actions, Parallel Computing, High-Performance Computing

**Publications** | *5 papers with 27 citations*

- *Microrobotic Locomotion in Blood Vessels: A Computational Study on the Performance...* Advanced Intelligent Systems, 2023.
- *The Mismatch between Experimental and CFD Analyses for Magnetic Surface Microrollers* Nature Scientific Reports, 2023.
- *Learning-Enhanced 3D Fiber Orientation Mapping in Thick Cardiac Tissues* Optica Open, 2025.
- *Anisotropic Surface Microrollers for Endovascular Navigation: Computational Analysis* Advanced Theory and Simulations, 2025.
- *Locomotion Behavior of Magnetic Microrollers in Confined Tubular Geometries...* Conference Paper, 2025.

## ACHIEVEMENTS & AWARDS

- Y Combinator AI Startup School (San Francisco, 2025): Selected among top 2000 CS students/grads globally.
- Viridien & Imperial Hackathon Winner: Developed ML models for Carbon Capture & Seismic Data Analysis.
- High School Entrance Exam: Ranked 1st/1M; University Entrance Exam: Ranked 300th/2M (Turkey); IELTS: 8.0.